Lecture 13: Basic Graph Algorithms

Michael Dinitz

October 15, 2026
601.433/633 Introduction to Algorithms

Introduction

Next 3-4 weeks: graphs!

- Super important abstractions, used all over the place in CS
  - Managing dependencies in software/project planning
  - Route planning
  - Network design/analysis
  - Image segmentation
- Great course on Graph Theory in AMS

Today: review of basic graph algorithms from Data Structures, possibly one or two new

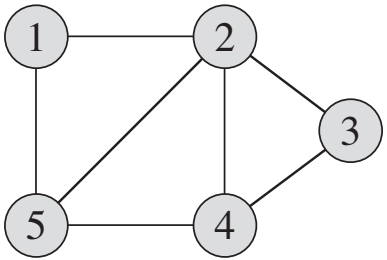
Basic Definitions

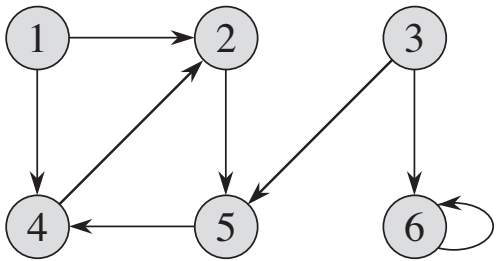
Definition

A graph $G = (V, E)$ is a pair where V is a set and $E \subseteq \binom{V}{2}$ (unordered pairs) or $E \subseteq V \times V$ (ordered pairs).

Notation:

- Elements of V are called *vertices* or *nodes*
- Elements of E are called *edges* or *arcs*.
- If $E \subseteq \binom{V}{2}$ then graph is *undirected*, if $E \subseteq V \times V$ graph is *directed*
- $|V| = n$ and $|E| = m$ (usually)
- So "size of input" $= n + m$





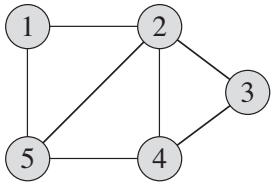
Representations

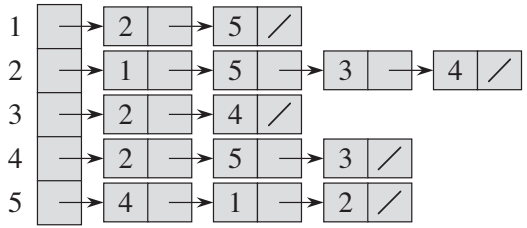
Adjacency List:

- Array A of length n
- $A[v]$ is linked list of vertices *adjacent* to v (edge from v to u)

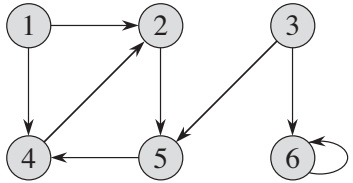
Adjacency Matrix:

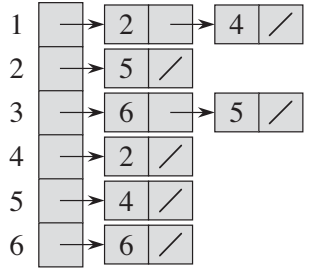
- $A \in \{0,1\}^{n \times n}$
- $A_{ij} = \begin{cases} 1 & \text{if } (i,j) \in E \\ 0 & \text{otherwise} \end{cases}$





	1	2	3	4	5
1	0	1	0	0	1
2	1	0	1	1	1
3	0	1	0	1	0
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- Replace adjacency *list* with adjacency *structure*: Red-black tree, hash table, etc.
- Not traditional, doesn't gain us much, and more complicated. But better!

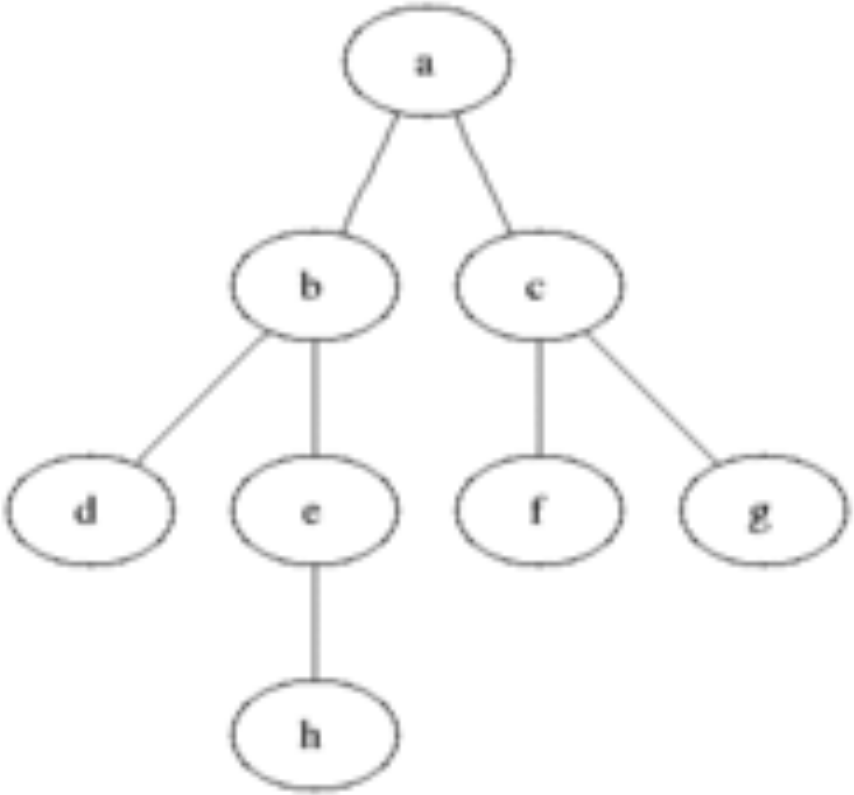
Breadth-First Search (BFS)

BFS Definition

Idea: explore graph in *levels* or *layers* from source s

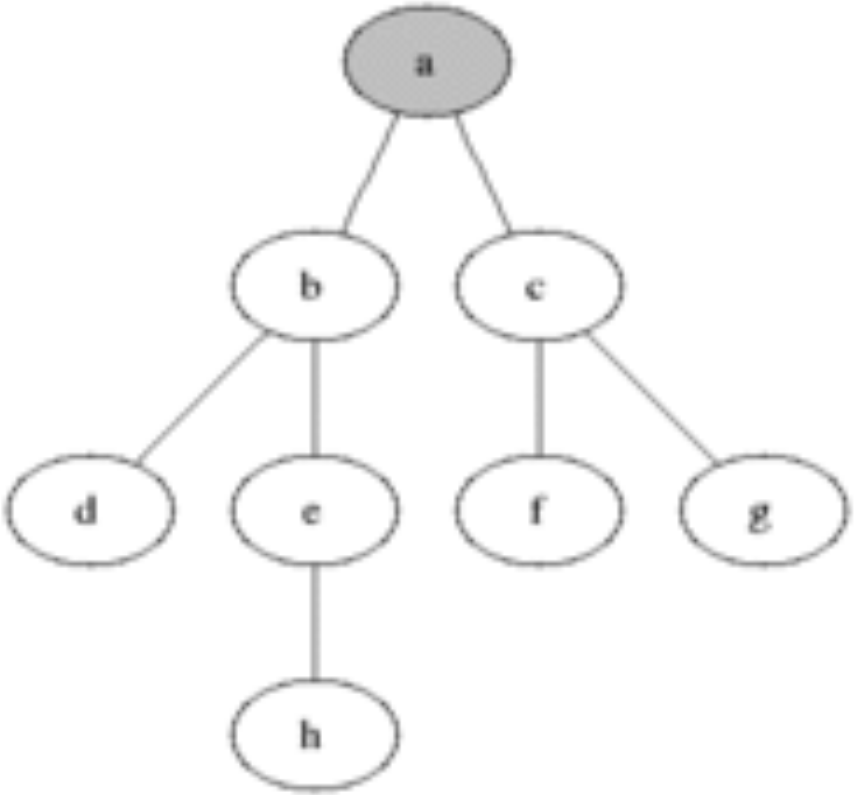
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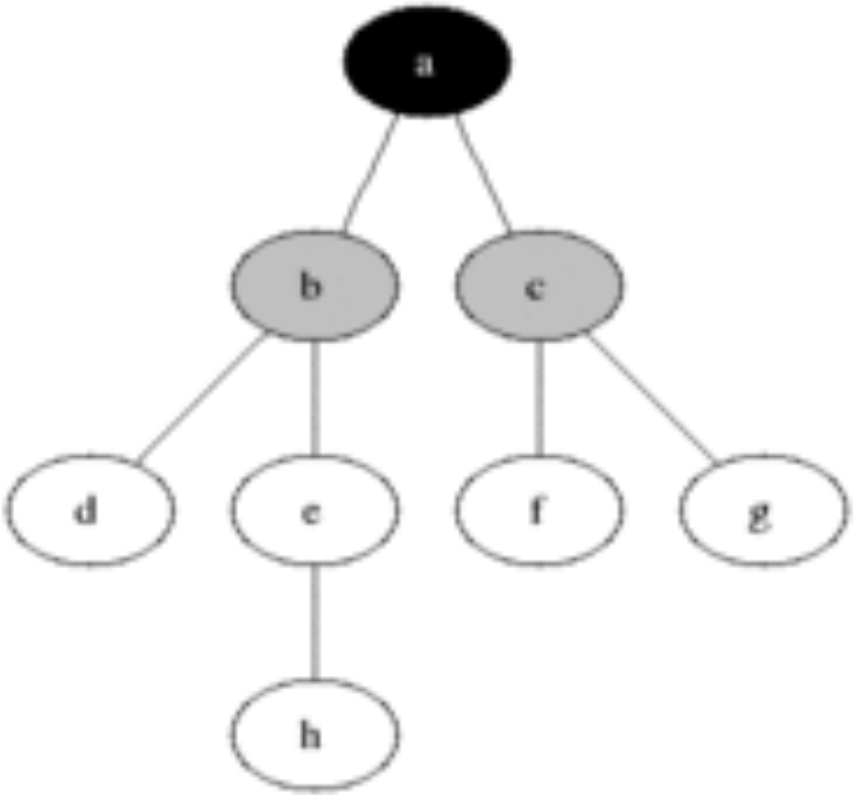
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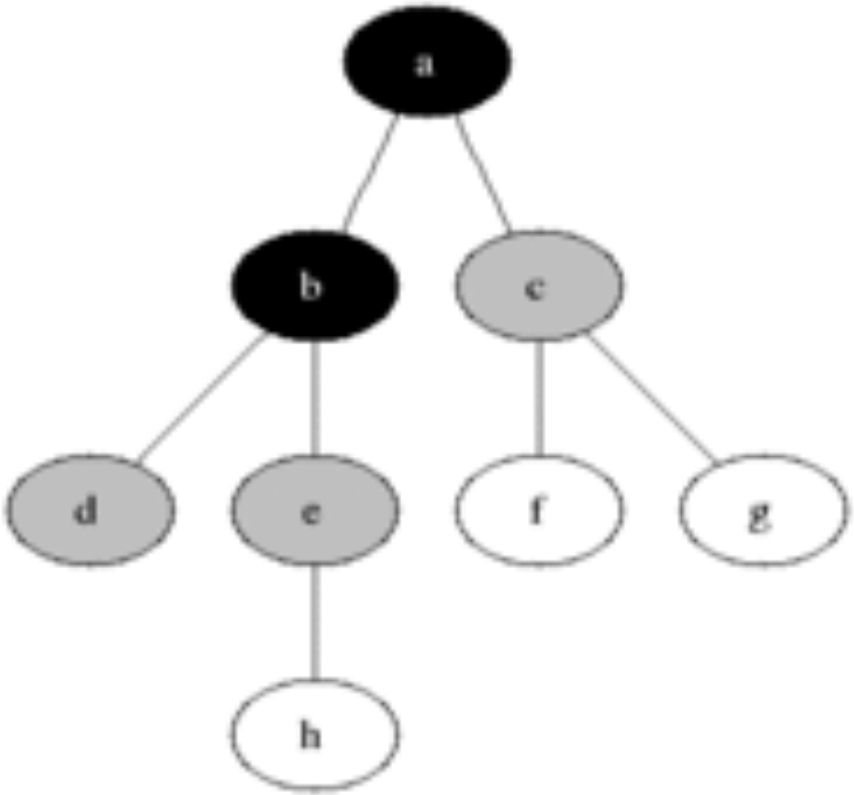
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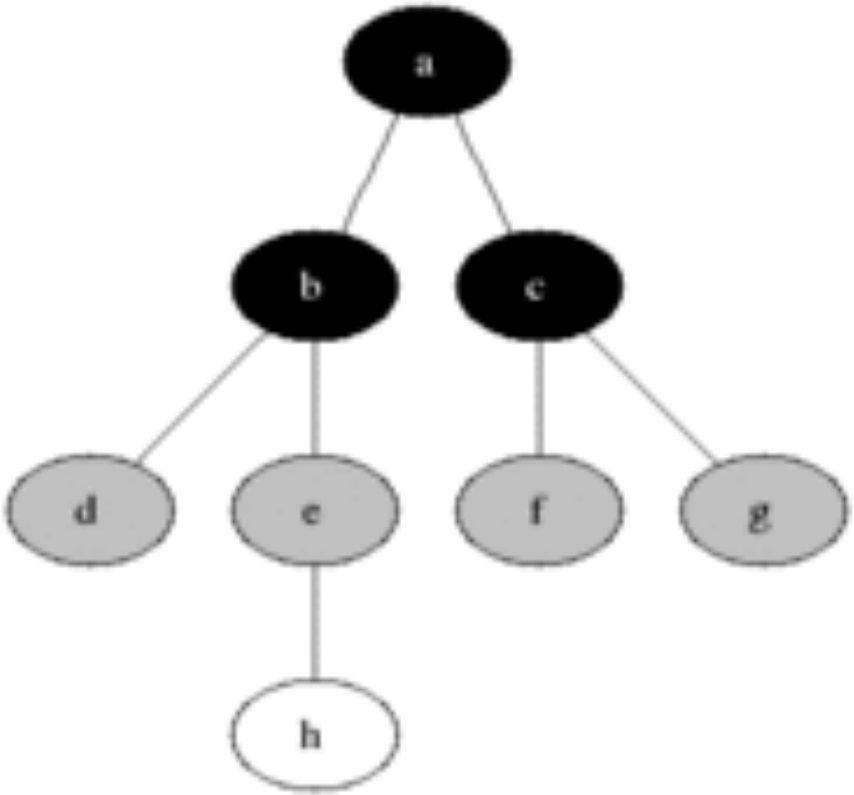
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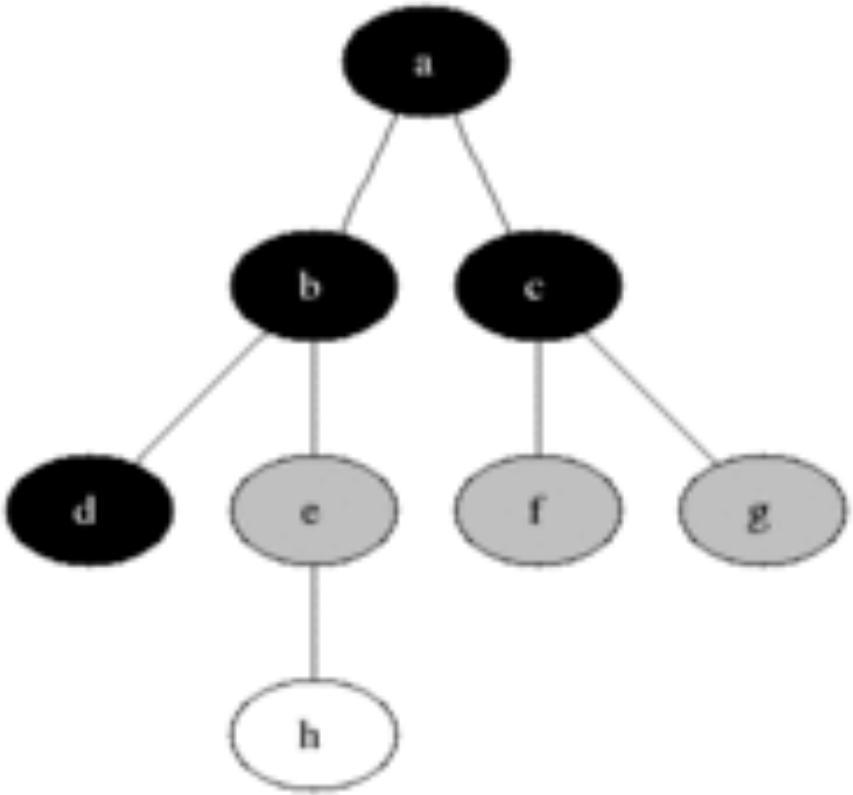
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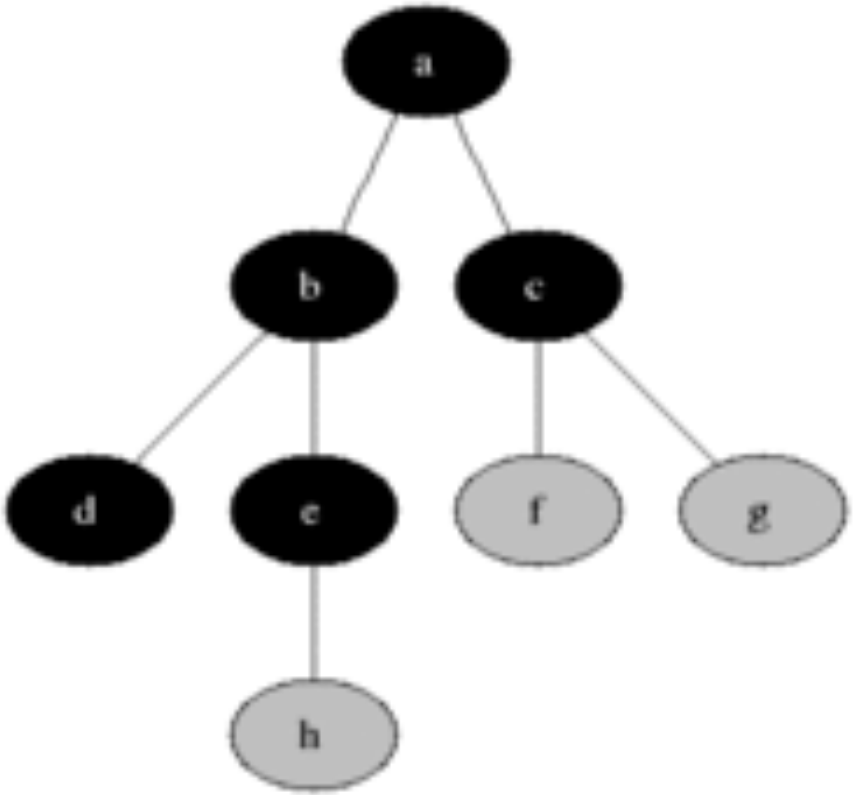
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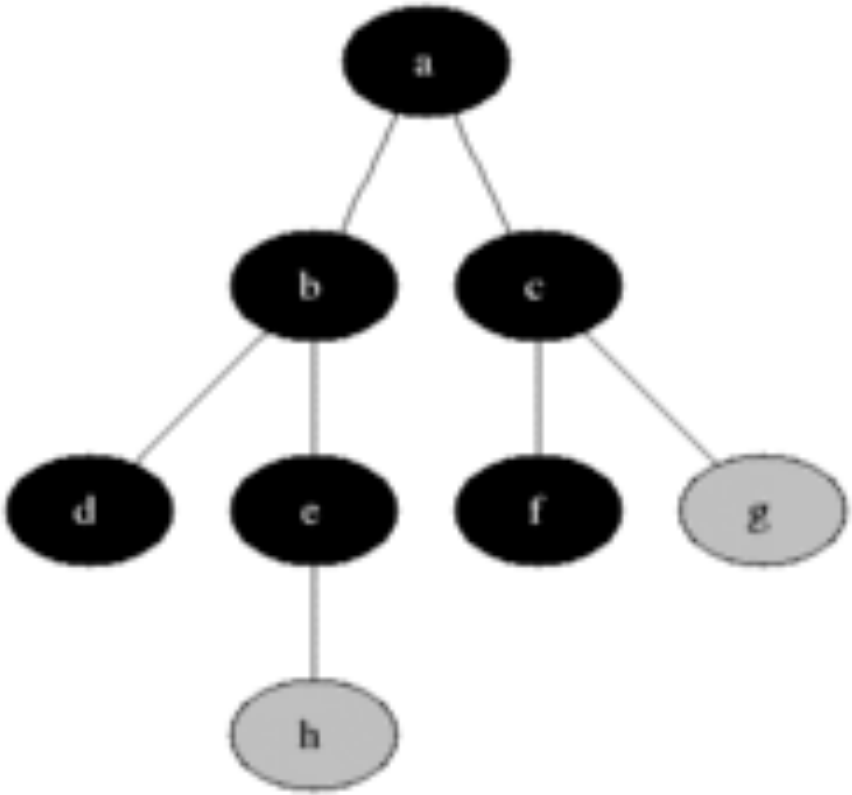
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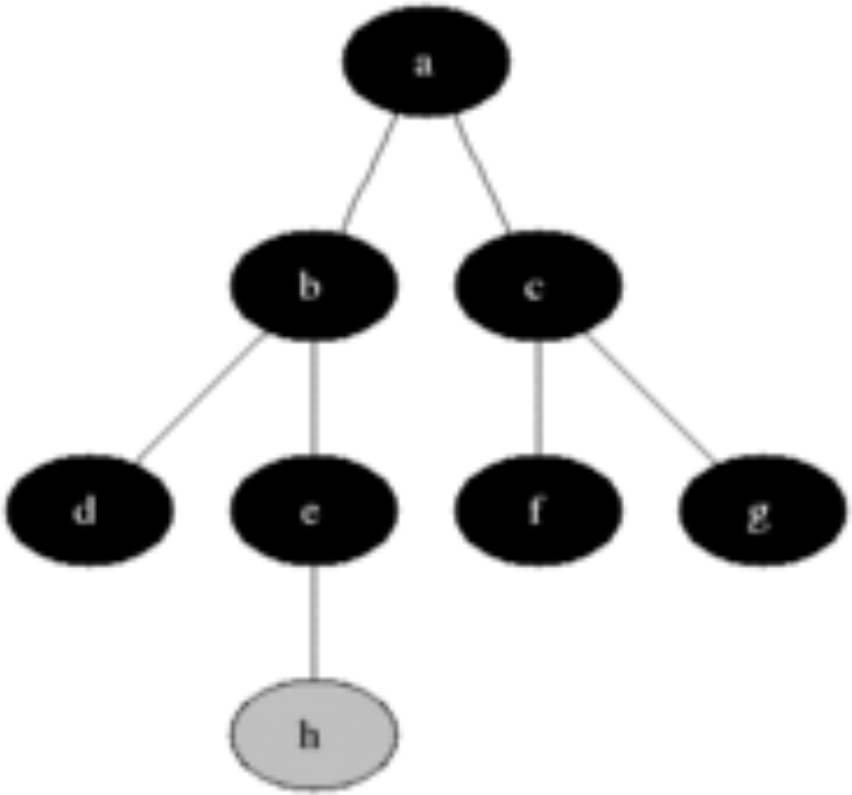
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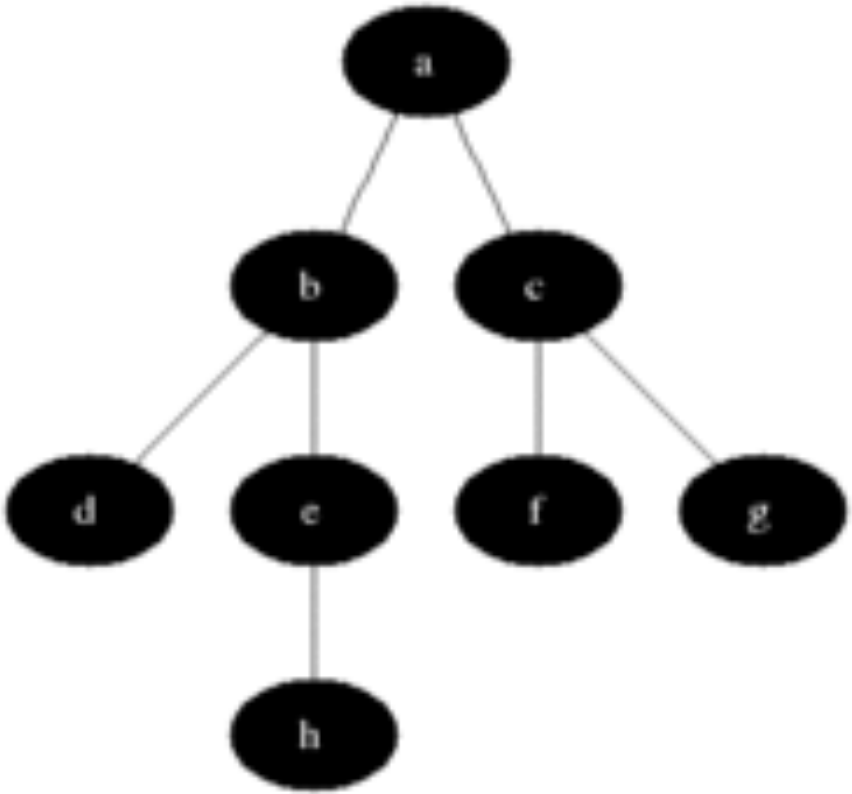
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BFS Pseudocode

Idea: explore with a queue (FIFO)

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BFS(G = (V, E), s) {
    Set mark(s) = True;
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    Enqueue(s);
    while(queue not empty) {
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Note: edges that cause a node to be enqueued form a tree!

Correctness / Shortest Paths

Definition: Distance $d(u, v)$ from u to v is min # edges in any path from u to v

Theorem (informal): BFS(s) gives shortest paths from s to all other nodes

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IH: $\forall j \leq i-1$, layer j of BFS tree $T = L_j$

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  $\implies \{u, v\} \in T \implies v$ at layer i of T.

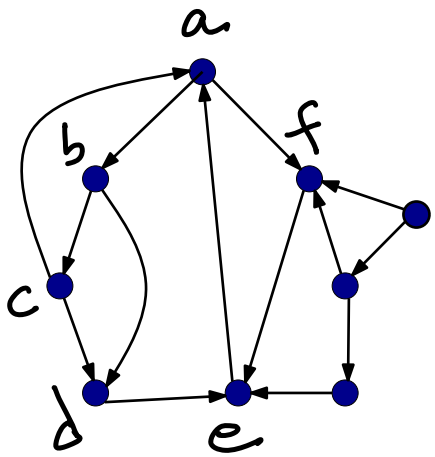
Depth-First Search (DFS)

DFS: Definition

Intuition: Instead of exploring wide (breadth), explore far (deep): just keep walking until see a node we've already seen, then backtrack!

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DFS(v) {
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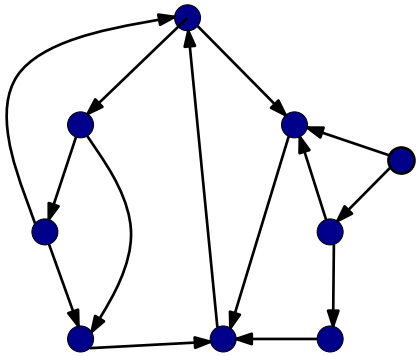


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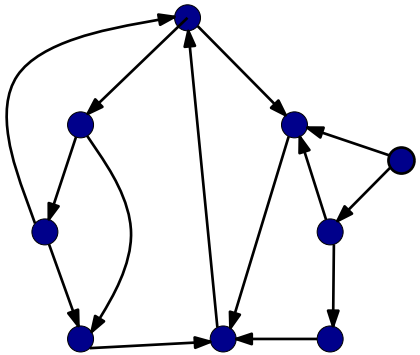
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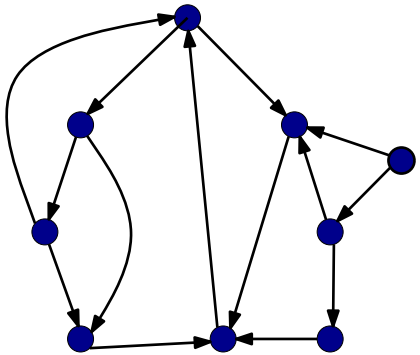
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Running time: $O(m+n)$

- $O(n)$ initialization
- Every edge considered at most twice

DFS: Correctness

Definition: u is *reachable* from v if there is a path $v = v_0, v_1, \ldots, v_k = u$ such that $(v_i, v_{i+1}) \in E$ for all $i \in \{0, 1, \ldots, k-1\}$.

Theorem

When DFS(v) terminates, it has visited (marked) all nodes that are reachable from v.

Proof.

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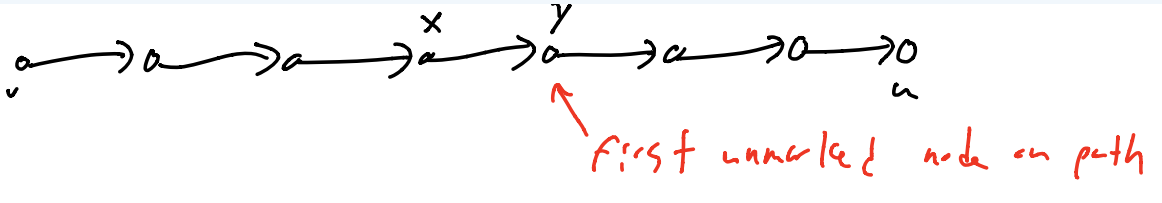
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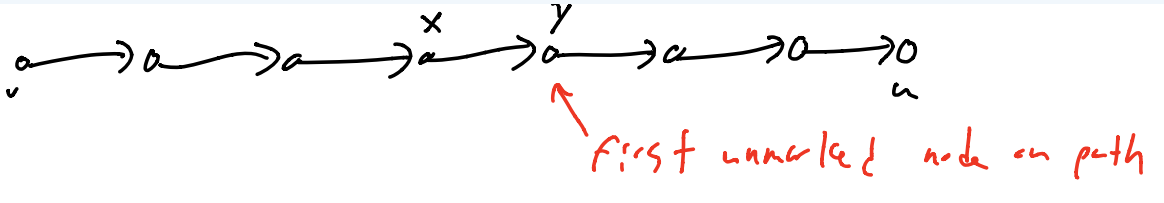
Definition: u is *reachable* from v if there is a path $v = v_0, v_1, \ldots, v_k = u$ such that $(v_i, v_{i+1}) \in E$ for all $i \in \{0, 1, \ldots, k-1\}$.

Theorem

When DFS(v) terminates, it has visited (marked) all nodes that are reachable from v.

Proof.

Suppose u reachable from v but not marked when DFS(v) terminates.



x is marked so DFS(x) must have been called

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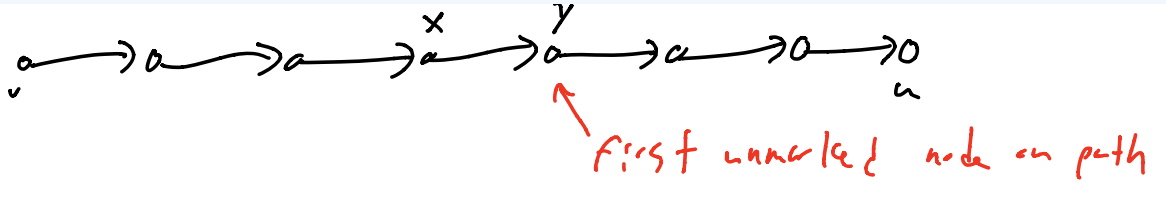
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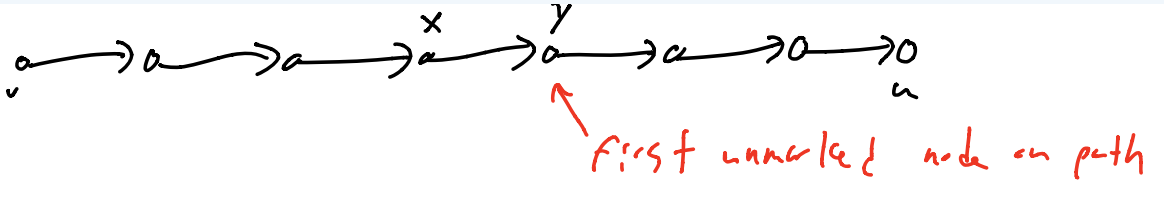
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Contradiction. □

Graph variant

After DFS($\boldsymbol{v}$), node marked if and only if reachable from $\boldsymbol{v}$.

Might want to continue until all nodes marked.

```
DFS(G) {
    for all v ∈ V, set mark(v) = False;
    while there exists an unmarked node v {
        DFS(v);
    }
}
```

Timestamps

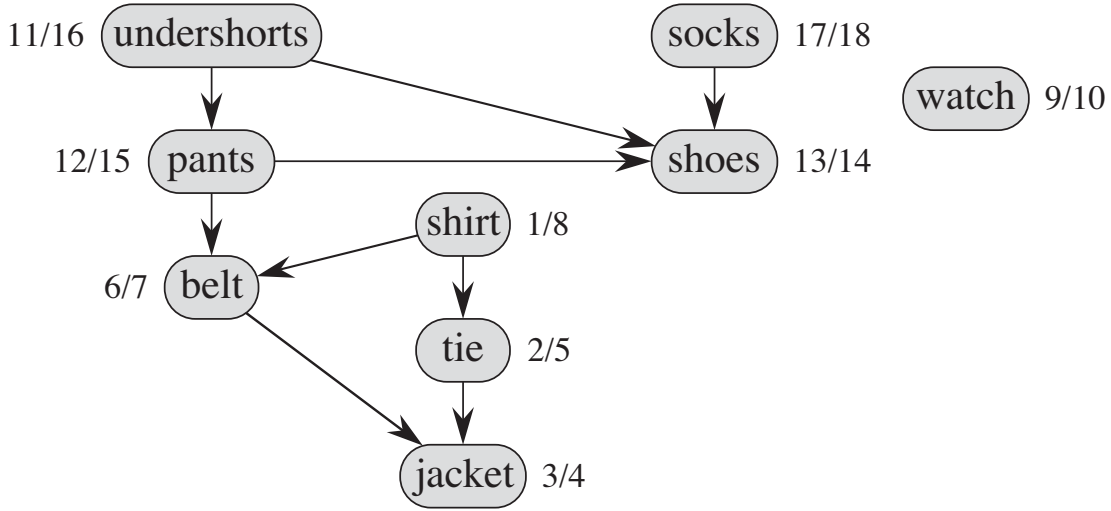
Explicitly keep track of "start" and "finishing" times

- Replaces ***mark***

```
DFS(G) {
    t = 0;
    for all v ∈ V {
        start(v) = 0;
        finish(v) = 0;
    }
    while ∃v ∈ V with start(v) = 0 {
        DFS(v);
    }
}
```

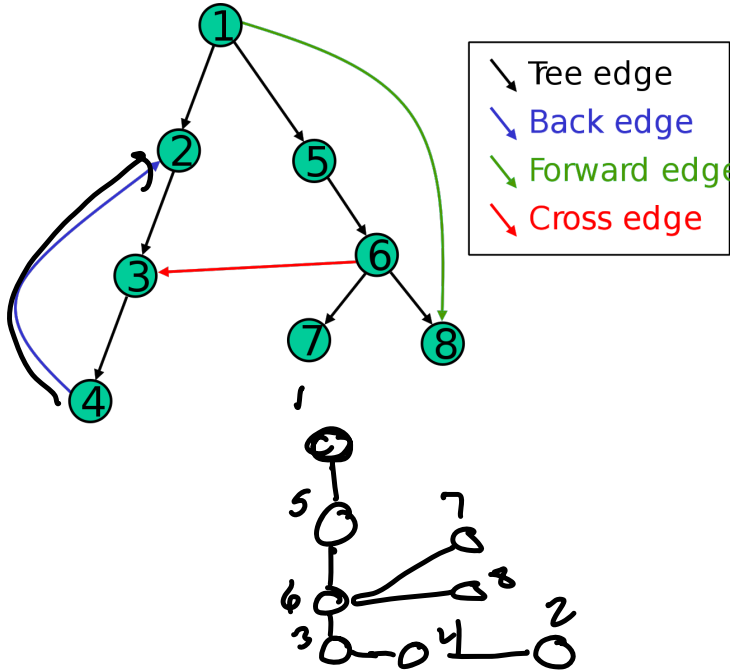
```
DFS(v) {
    t = t + 1;
    start(v) = t;
    for each edge (v, u) ∈ A[v] {
        if start(u) == 0 then DFS(u);
    }
    t = t + 1;
    finish(v) = t;
}
```

Timestamp Example



Edge Types

DFS naturally gives a spanning forest: edge (v, u) if DFS(v) calls DFS(u)



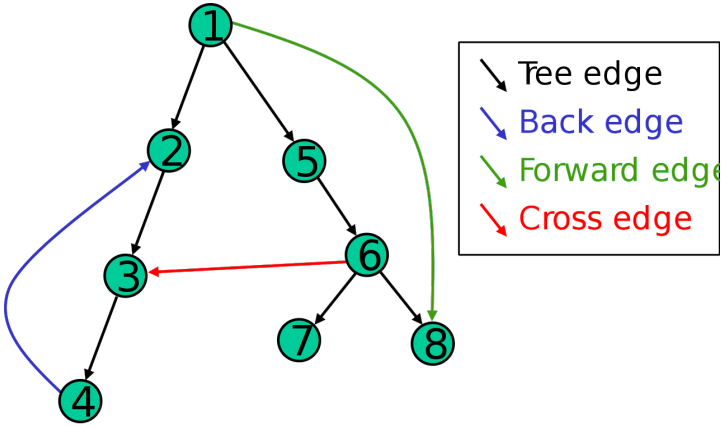
Forward Edges: (v, u) such that u descendant of v (includes tree edges)

Back Edges: (v, u) such that u an ancestor of v

Cross Edges: (v, u) such that u neither a descendant nor an ancestor of v

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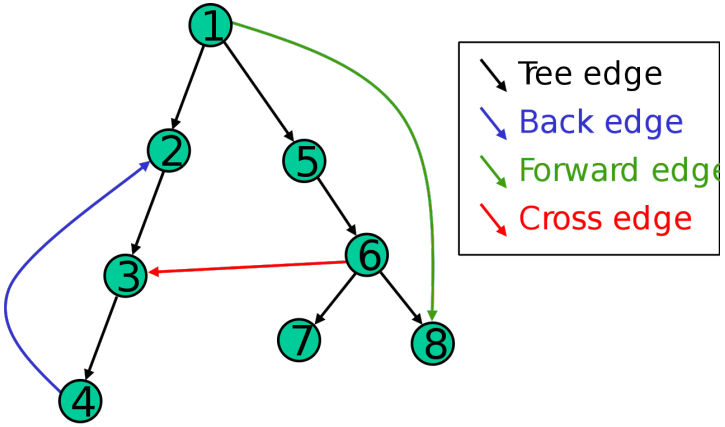
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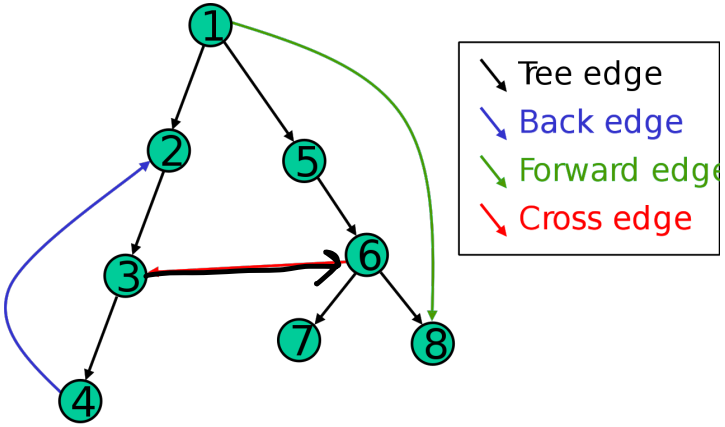
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